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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Report

This report develops a regime-based allocation approach that rotates among TLT, GLD, and SPY using the volatility setting inferred from VIX moves. The last implementation ends up using a 3 state Gaussian Hidden Markov Model, and it adds a 1-day execution lag to reduce look-ahead bias. Then it compares everything versus buy and hold SPY, and also against a monthly rebalanced equal weight benchmark, as the project brief asked for.

From the final results, the preferred implementation is a 3-state framework called Calm, Moderate, and Turbulent. The strategy sends 100% into whichever ETF shows the highest historical mean return inside the current regime. It also manages to show stronger total return, and a higher Sharpe ratio than buy-and-hold SPY through the sample. Meanwhile the equal weight benchmark stays pretty competitive, at least from a risk adjusted point of view.

1. Data Preparation and Exploration

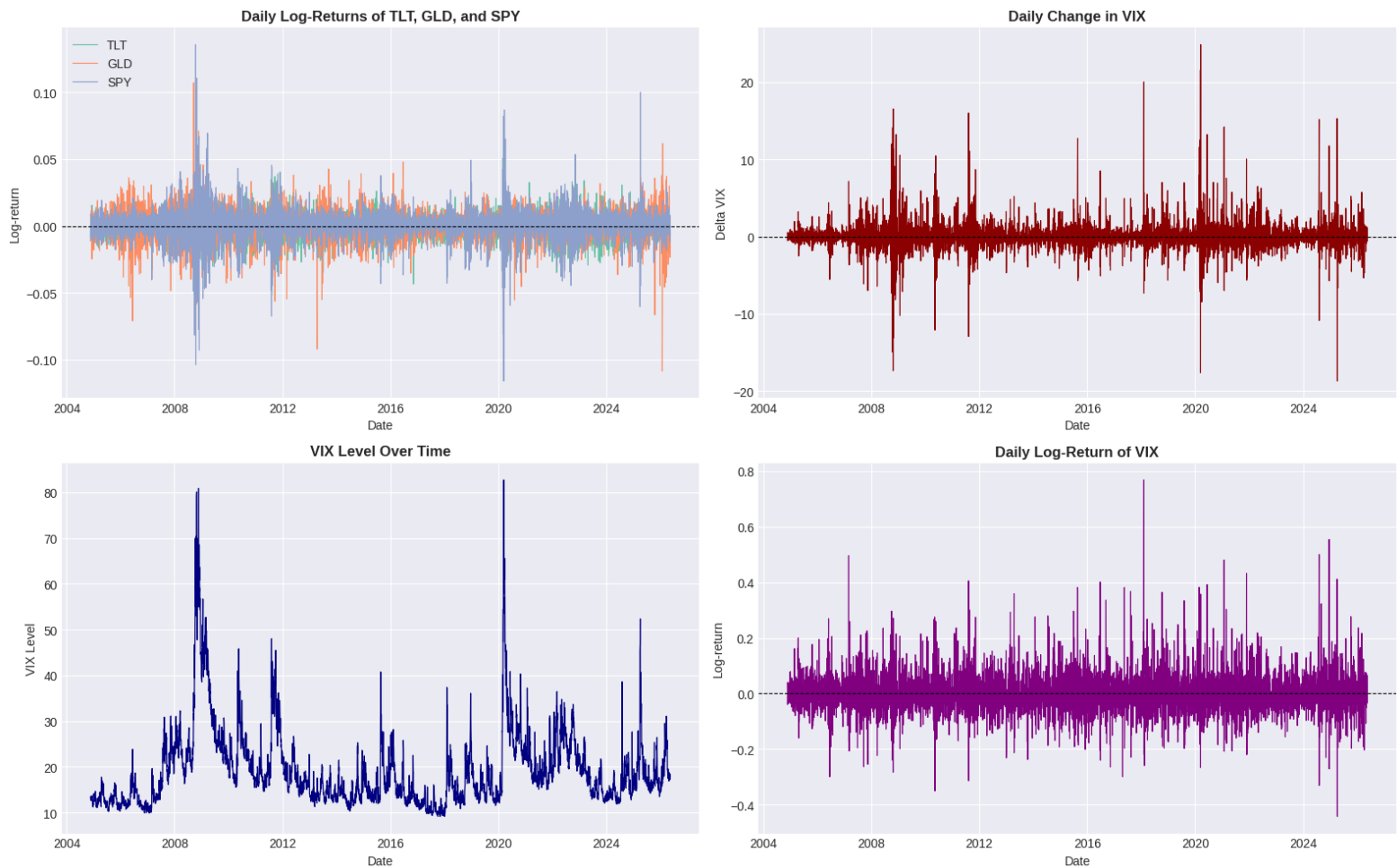
The analysis starts with daily adjusted close prices for TLT, GLD, SPY, and VIX that were downloaded from Yahoo Finance across the maximum overlapping time window. After we align the series and drop missing values, the working dataset ends up with 5,407 daily observations, covering 2004-11-19 to 2026-05-19. For returns, log returns are computed for the three ETFs, and for VIX we compute daily changes.

This preparation step matters, because the regime model needs a common trading calendar across all assets and also the volatility proxy. Daily VIX change was chosen as the state variable, since it reflects shifts in market unease more directly than the static VIX level. Also, it is the exact input later used inside the Hidden Markov Model.

GROUP WORK PROJECT # 2

Group Number: 14881

MScFE 622: STOCHASTIC MODELING

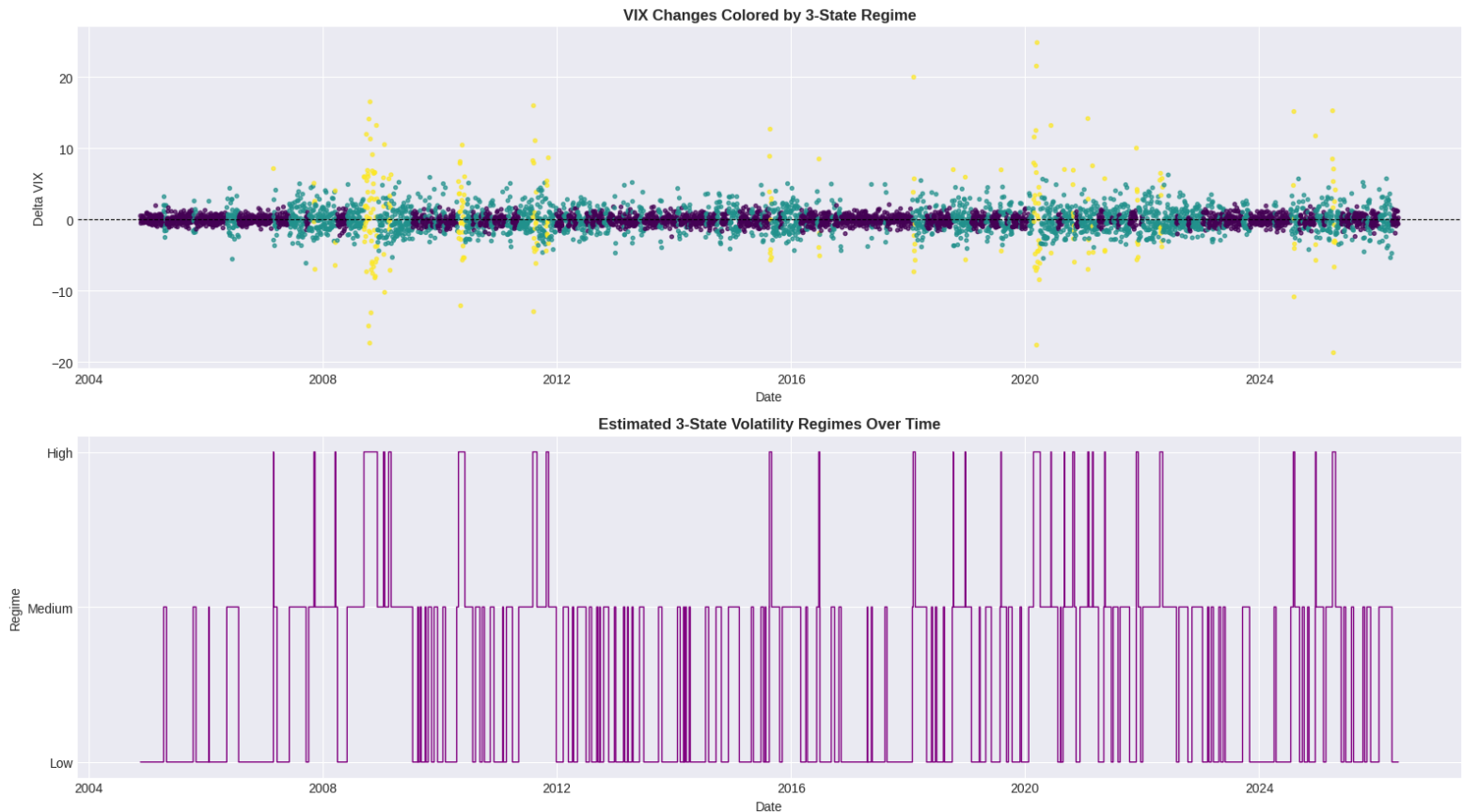


Exploratory statistics already sort of suggest that the three assets react differently when volatility regimes shift. SPY looks like it has the biggest downside sensitivity when things get really stressed, and TLT plus GLD tend to offer diversification potential, so overall it feels like a natural backdrop for a regime-based rotation idea.

2. Modelling VIX Regimes

We used the Hidden Markov Model option. Two baseline Gaussian HMMs were fitted using daily VIX changes as the observation series, and specifically, the required 2 state and 3 state variants were used.

The 2-state model yields a log-likelihood of -9210.09, with an AIC of 18436.18, and a BIC of 18488.94. The 3-state model improves the fit quite a bit, giving a log-likelihood of -8865.64, an AIC of 17761.28, and a BIC of 17860.21, which implies the extra state actually grabs something important about volatility dynamics, not just noise.



The main conclusion from this step is that the 3-state model is sufficiently rich to separate calm, transitional, and turbulent volatility environments. This gives a stable operational framework without immediately overcomplicating the allocation rule.

3. State Selection and Interpretation

To evaluate model granularity, we extended the comparison beyond the required 2- and 3-state HMMs and also fits 4-state and 5-state versions as sensitivity checks. The full comparison table shows that information criteria continue to decline as states are added: the 4-state model has AIC 17554.07 and BIC 17712.36, while the 5-state model has AIC 17433.74 and BIC 17664.58, making the 5-state specification the statistical winner by both AIC and BIC.

Despite this, the final implementation deliberately selects the 3-state Gaussian HMM. The 3-state structure preserves interpretability, reduces the risk of overfitting, and aligns directly with a usable market narrative of calm, moderate, and stressed volatility regimes.

The ordered state summaries for the selected model are as follows. State 0 is labelled Low Volatility (Calm), with mean VIX change -0.0600 and variance 0.3851; State 1 is Medium Volatility (Moderate), with mean VIX change -0.0128 and variance 3.1003; State 2 is High Volatility (Turbulent), with mean VIX change 0.6511 and variance 36.3550.

The asset behaviour by state provides the economic foundation for the strategy. In the Calm state, SPY has the highest mean daily return at 0.001301, compared with 0.000428 for GLD and -0.000159 for TLT. In the Moderate state, GLD has the highest mean daily return at 0.000441, slightly above TLT at 0.000295 and above SPY at approximately zero. In the Turbulent state, TLT dominates with a mean daily return of 0.001556, while GLD is slightly negative at -0.000015 and SPY performs worst at -0.005700.

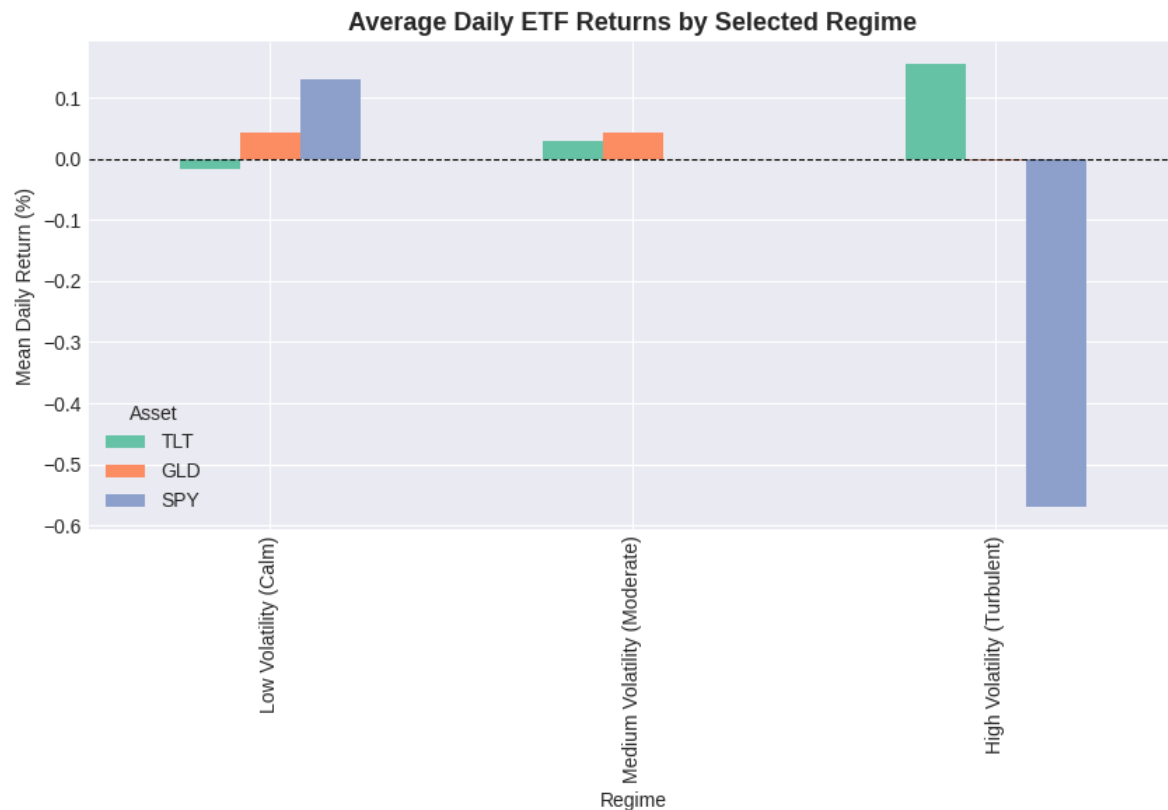
These results imply a coherent economic interpretation. Equities perform best in calm regimes, gold becomes relatively attractive in intermediate conditions, and long Treasuries dominate in stress episodes when equity losses are most severe.

Model Comparison Table

Model	Log-Likelihood	AIC	BIC	Model	Log-Likelihood
2-state HMM	-9210.09	18436.18	18488.94	2-state HMM	-9210.09
3-state HMM	-8865.64	17761.28	17860.21	3-state HMM	-8865.64
4-state HMM	-8753.04	17554.07	17712.36	4-state HMM	-8753.04
5-state HMM	-8681.87	17433.74	17664.58	5-state HMM	-8681.87

State-Conditional Return Table for Selected 3-State Model

State	Label	Count	TLT Mean Return	TLT Std. Dev.	GLD Mean Return	GLD Std. Dev.	SPY Mean Return	SPY Std. Dev.
0	Low Volatility (Calm)	2853	-0.000159	0.007452	0.000428	0.009445	0.001301	0.005604
1	Medium Volatility (Moderate)	2292	0.000295	0.009644	0.000441	0.012196	-0.000001	0.012341
2	High Volatility (Turbulent)	262	0.001556	0.017916	-0.000015	0.020719	-0.005	



The final model choice is therefore the 3-state Gaussian HMM, not because it minimizes AIC or BIC, but because it offers the best compromise between statistical fit, interpretability, and usability in a rule-based allocation framework. This directly answers the project requirement to justify the preferred number of states and explain the investment relevance of the selected regime structure.

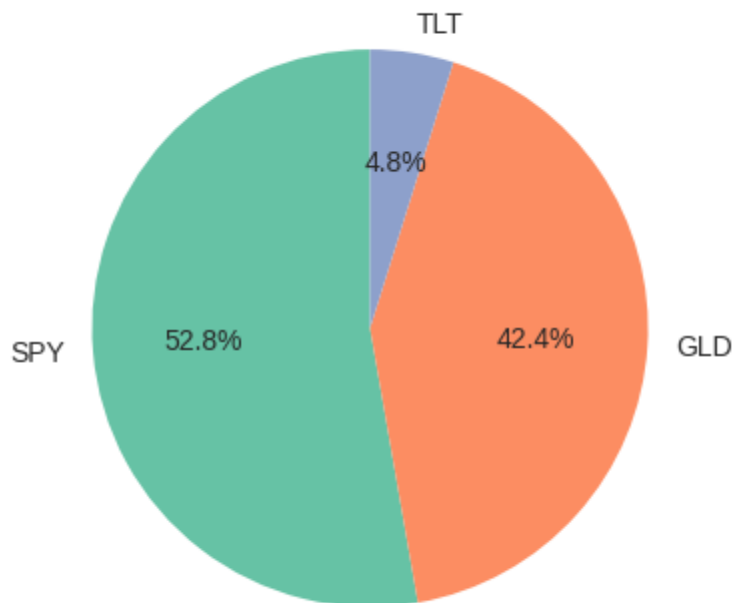
4. Designing the Rotation Strategy

The allocation rule follows the project instructions directly: for each state, the portfolio allocates 100% to the ETF with the highest historical mean return in that state. A 1-day execution lag is introduced so that the state inferred on date t determines the position held on date $t + 1$, reducing look-ahead bias and making the implementation more realistic. Based on the selected 3-state model, the final allocation map is straightforward. In Calm regimes the strategy allocates to SPY, in Moderate regimes it allocates to GLD, and in turbulent regimes it allocates to TLT.

Final Allocation Mapping

State	Label	Allocate 100% To
0	Low Volatility (Calm)	SPY
1	Medium Volatility (Moderate)	GLD
2	High Volatility (Turbulent)	TLT

This rule can be interpreted as an automated flight-to-safety policy. When volatility is low, the strategy accepts equity risk to capture stronger upside; when volatility rises to transitional levels, it shifts into gold; and when volatility becomes turbulent, it rotates into long-duration Treasuries, which historically performed best in the selected stress regime.

Distribution of Chosen Assets in Strategy

5. Backtesting and Performance Evaluation

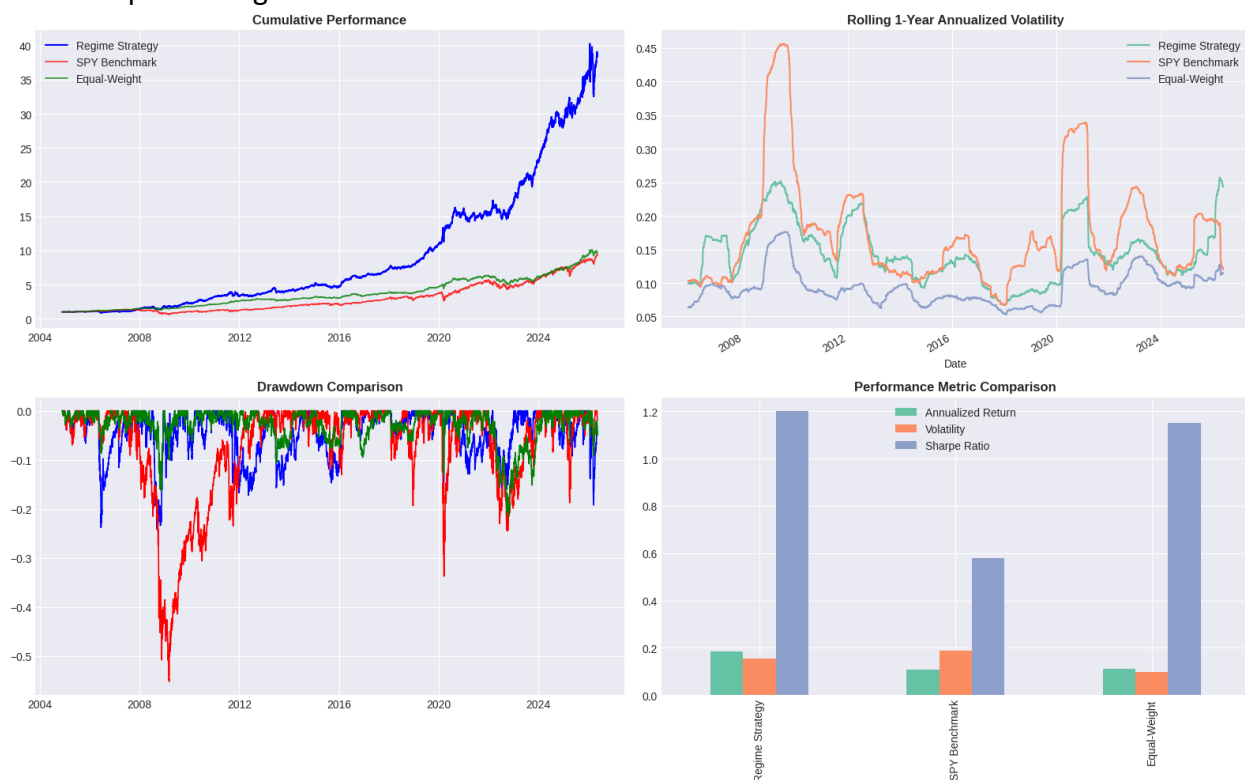
The backtest, compares the regime approach against two required comparison points, basically buy and hold SPY , plus an equal-weight mix across TLT, GLD, and SPY. The performance is summarized with cumulative return, annualized return, volatility, Sharpe ratio, and maximum drawdown, so it aligns with the evaluation criteria.

Performance Summary

Portfolio	Cumulative Return	Annualized Return	Volatility	Sharpe Ratio	Max Drawdown
Regime Strategy	37.3150	0.1852	0.1545	1.1992	-0.2401
SPY Benchmark	8.2804	0.1094	0.1894	0.5778	-0.5519
Equal-Weight Benchmark	8.6355	0.1114	0.0968	1.1502	-0.2160

Overall, the regime strategy clearly beats the buy-and-hold SPY benchmark in both the plain return sense and the risk adjusted picture. It posts a higher cumulative result, also a higher annualized return, it shows noticeably reduced volatility, a much stronger Sharpe ratio, and a meaningfully less severe maximum drawdown compared to the passive equity route.

Compared with the equal-weight benchmark, the regime strategy still looks better in absolute terms, and it also has a slightly improved Sharpe ratio. However, the equal-weight portfolio ends up with lower volatility, and it also shows a marginally smaller maximum drawdown. This means the regime strategy appears to offer the best growth profile among the three portfolios, while the equal-weight benchmark remains a strong diversified alternative for investors prioritizing steadier risk control.



The preferred 3-state model appears to be the most stable and communicable choice for implementation. A higher-state model may improve in-sample fit, but it would also increase strategy complexity and reduce interpretability, which is a meaningful drawback in a practical investment policy designed for transparent rule-based execution.

6. Practical Recommendation

The final recommendation is to adopt the 3-state regime-based allocation framework as the preferred strategy from this project. It is transparent, economically intuitive, and clearly superior to buy-and-hold SPY on both return and downside risk over the historical sample.

The recommended operating rule is simple: hold SPY in calm volatility regimes, rotate to GLD in moderate regimes, and shift to TLT in turbulent regimes, always with a 1-day execution lag. This approach preserves the practical objective of adapting portfolio exposure to the market's changing volatility environment while avoiding discretionary decision-making.

At the same time, the historical results should be interpreted with appropriate caution. The strategy relies on in-sample state-conditional averages, so future performance will depend on whether regime behaviour remains similar out of sample; therefore, the rule should be viewed as a disciplined evidence-based allocation policy rather than a guarantee of superior future returns.

References

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